CLASS PARTICIPATION

MACHINE LEARNING-MDA2407

ACADEMIC YEAR(2025-2027)

FAKE NEWS DETECTION USING LOGISTIC REGRESSOR

DONE BY

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INTRODUCTION

The rapid growth of the internet and social media platforms has significantly increased the speed at which information spreads across the world. While this has improved communication and access to information, it has also led to the widespread dissemination of fake or misleading news. Fake news refers to false or fabricated information presented as legitimate news, often intended to mislead readers or influence public opinion. The spread of fake news can create confusion, manipulate political opinions, damage reputations, and cause social unrest. Detecting such misinformation manually is difficult due to the massive amount of content generated every day. Therefore, automated systems using machine learning and natural language processing techniques are essential for identifying and filtering fake news articles. This project aims to build a machine learning model that can classify news articles as real or fake based on textual features extracted from the news content. By analysing patterns in the language, structure, and content of news articles, the model helps in identifying misinformation and improving the reliability of information shared online.

USE CASE

Fake news detection systems are widely used by social media platforms, news organizations, and fact-checking agencies to identify misleading information before it spreads widely. With the increasing use of digital platforms for news consumption.Fake news detection use cases automated detection systems can help maintain the credibility of online information sources. In this use case, a machine learning model is developed to analyze news articles and determine whether they are genuine or fake. By examining the textual content of the articles, the system can identify patterns commonly associated with misinformation. This model can be integrated into content moderation systems to flag suspicious news articles and assist journalists, researchers, and social media platforms in verifying the authenticity of news content. Such systems can reduce the spread of misinformation, protect users from misleading information, and promote a more trustworthy digital information environment

PROBLEM STATEMENT

The rapid spread of fake news on online platforms has become a major challenge in the digital information age. Misleading news articles can influence public opinion, create panic, and negatively impact social, political, and economic stability. Manually verifying every piece of news

content is impractical due to the vast amount of information shared daily across the internet. This project aims to develop a machine learning-based fake news detection model that can automatically classify news articles as real or fake. By utilizing natural language processing techniques and machine learning algorithms such as Logistic Regression, decision tree, the model analyzes textual patterns within news articles to detect misinformation.

The objective is to improve the accuracy of fake news identification and provide a reliable tool that helps users, media organizations, and online platforms detect and limit the spread of false information

JUPYTER NOTEBOOK

Jupyter Notebook is an open-source, web-based interactive computing environment that allows users to create and share documents containing live code, equations, visualizations, and narrative text. It supports multiple programming languages, with Python being the most common, and is widely used in data science, machine learning, and scientific computing for exploratory analysis, visualization, and prototyping. The name "Jupyter" is derived from the three core programming languages it originally supported: Julia, Python, and R.

ALGORITHMS USED

1.Logistic Regressor.

DATASET DESCRIPTION.

Dataset:" https://www.kaggle.com/datasets/jainpooja/fake-news-detection?select=Fake.csv"

"https://www.kaggle.com/datasets/jainpooja/fake-news-detection?select=True.csv"

1.Title - The headline of the news article. It gives a short summary of the news.

2.Text - The full content/body of the news article. This is the main text used for analysis in fake news detection.

3.Subject - The category or the topic of the news (for example :Politics ,world news ,etc..)

4.Date – The date when the news article was published

	A	B	C	D	E
1	**TITLE**	**TEXT**	**SUBJECT**	**DATE**	
2	As U.S. budget fight looms, Republicans flip th	WASHINGTON (Reuters) - The head of a conse	politicsNews	31-Dec-17	
3	U.S. military to accept transgender recruits on	WASHINGTON (Reuters) - Transgender people	politicsNews	29-Dec-17	
4	Senior U.S. Republican senator: 'Let Mr. Muell	WASHINGTON (Reuters) - The special counse	politicsNews	31-Dec-17	
5	FBI Russia probe helped by Australian diploma	WASHINGTON (Reuters) - Trump campaign ac	politicsNews	30-Dec-17	
6	Trump wants Postal Service to charge 'much n	SEATTLE/WASHINGTON (Reuters) - President	politicsNews	29-Dec-17	
7	White House, Congress prepare for talks on sp	WEST PALM BEACH, Fla./WASHINGTON (Reut	politicsNews	29-Dec-17	
8	Trump says Russia probe will be fair, but timeli	WEST PALM BEACH, Fla (Reuters) - President	politicsNews	29-Dec-17	
9	Factbox: Trump on Twitter (Dec 29) - Approval	The following statementsÂ were posted to the	politicsNews	29-Dec-17	
10	Trump on Twitter (Dec 28) - Global Warming	The following statementsÂ were posted to the	politicsNews	29-Dec-17	
11	Alabama official to certify Senator-elect Jones	WASHINGTON (Reuters) - Alabama Secretary	politicsNews	28-Dec-17	
12	Jones certified U.S. Senate winner despite Mo	(Reuters) - Alabama officials on Thursday certi	politicsNews	28-Dec-17	
13	New York governor questions the constitution	NEW YORK/WASHINGTON (Reuters) - The nev	politicsNews	28-Dec-17	
14	Factbox: Trump on Twitter (Dec 28) - Vanity Fa	The following statementsÂ were posted to the	politicsNews	28-Dec-17	
15	Trump on Twitter (Dec 27) - Trump, Iraq, Syria	The following statementsÂ were posted to the	politicsNews	28-Dec-17	
16	Man says he delivered manure to Mnuchin to p	(In Dec. 25 story, in second paragraph, correc	politicsNews	25-Dec-17	
17	Virginia officials postpone lottery drawing to de	(Reuters) - A lottery drawing to settle a tied Virg	politicsNews	27-Dec-17	
18	U.S. lawmakers question businessman at 201	WASHINGTON (Reuters) - A Georgian-Americ	politicsNews	27-Dec-17	
19	Trump on Twitter (Dec 26) - Hillary Clinton, Tax	The following statementsÂ were posted to the	politicsNews	26-Dec-17	
20	U.S. appeals court rejects challenge to Trump	(Reuters) - A U.S. appeals court in Washington	politicsNews	26-Dec-17	
21	Treasury Secretary Mnuchin was sent gift-wra	(Reuters) - A gift-wrapped package addressed	politicsNews	24-Dec-17	
22	Federal judge partially lifts Trump's latest refu	WASHINGTON (Reuters) - A federal judge in Se	politicsNews	24-Dec-17	
23	Exclusive: U.S. memo weakens guidelines for	NEW YORK (Reuters) - The U.S. Justice Depart	politicsNews	23-Dec-17	
24	Trump travel ban should not apply to people wi	(Reuters) - A U.S. appeals court on Friday said	politicsNews	23-Dec-17	
25	Second court rejects Trump bid to stop transge	WASHINGTON (Reuters) - A federal appeals c	politicsNews	23-Dec-17	
26	Failed vote to oust president shakes up Peru's	LIMA (Reuters) - Peruâ€™s President Pedro Pa	politicsNews	23-Dec-17	

True

	A	B	C	D
1	**TITLE**	**TEXT**	**SUBJECT**	**DATE**
2	Donald Trump Sends Out Embarrassing	Donald Trump just couldn t wish all Americans a Hap	News	31-Dec-17
3	Drunk Bragging Trump Staffer Started R	House Intelligence Committee Chairman Devin Nun	News	31-Dec-17
4	Sheriff David Clarke Becomes An Intern	On Friday, it was revealed that former Milwaukee Sh	News	30-Dec-17
5	Trump Is So Obsessed He Even Has Ob	On Christmas day, Donald Trump announced that he	News	29-Dec-17
6	Pope Francis Just Called Out Donald Tr	Pope Francis used his annual Christmas Day messa	News	25-Dec-17
7	Racist Alabama Cops Brutalize Black B	The number of cases of cops brutalizing and killing p	News	25-Dec-17
8	Fresh Off The Golf Course, Trump Lash	Donald Trump spent a good portion of his day at his g	News	23-Dec-17
9	Trump Said Some INSANELY Racist Stu	In the wake of yet another court decision that deraile	News	23-Dec-17
10	Former CIA Director Slams Trump Over	Many people have raised the alarm regarding the fac	News	22-Dec-17
11	WATCH: Brand-New Pro-Trump Ad Fea	Just when you might have thought we d get a break fr	News	21-Dec-17
12	Papa Johnâ€™s Founder Retires, Figure	A centerpiece of Donald Trump s campaign, and nov	News	21-Dec-17
13	WATCH: Paul Ryan Just Told Us He Doe	Republicans are working overtime trying to sell their	News	21-Dec-17
14	Bad News For Trump â€" Mitch McCon	Republicans have had seven years to come up with a	News	21-Dec-17
15	WATCH: Lindsey Graham Trashes Medi	The media has been talking all day about Trump and	News	20-Dec-17
16	Heiress To Disney Empire Knows GOP S	Abigail Disney is an heiress with brass ovaries who w	News	20-Dec-17
17	Tone Deaf Trump: Congrats Rep. Scalis	Donald Trump just signed the GOP tax scam into law	News	20-Dec-17
18	The Internet Brutally Mocks Disneyâ€™s	A new animatronic figure in the Hall of Presidents at	News	19-Dec-17
19	Mueller Spokesman Just F-cked Up Dor	Trump supporters and the so-called president s favo	News	17-Dec-17
20	SNL Hilariously Mocks Accused Child M	Right now, the whole world is looking at the shocking	News	17-Dec-17
21	Republican Senator Gets Dragged For C	Senate Majority Whip John Cornyn (R-TX) thought it w	News	16-Dec-17
22	In A Heartless Rebuke To Victims, Trum	It almost seems like Donald Trump is trolling Americ	News	16-Dec-17
23	KY GOP State Rep. Commits Suicide Ov	In this #METOO moment, many powerful men are be	News	13-Dec-17
24	Meghan McCain Tweets The Most AMAZ	As a Democrat won a Senate seat in deep-red Alaba	News	12-Dec-17
25	CNN CALLS IT: A Democrat Will Repres	Alabama is a notoriously deep red state. It s a place	News	12-Dec-17
26	White House: It Wasnâ€™t Sexist For Tr	A backlash ensued after Donald Trump launched a s	News	12-Dec-17

Fake

LOGISTIC REGRESSOR

Logistic Regression is a supervised machine learning algorithm used for **classification problems**. It predicts the **probability of a categorical outcome**, such as yes/no or true/false. The algorithm uses the **logistic (sigmoid) function** to convert predicted values into probabilities between 0 and 1. It works well for **binary classification tasks** like spam detection or disease prediction. Logistic Regression is simple, fast, and easy to interpret. However, it performs best when the relationship between variables is **linear**.

SOURCE CODE

```
#import necessary modules

import pandas as pd

import numpy as np

import seaborn as sns

import matplotlib.pyplot as plt

from sklearn.feature_extraction.text import TfidfVectorizer

from sklearn.model_selection import train_test_split

from sklearn.linear_model import LogisticRegression

from sklearn.metrics import accuracy_score

from sklearn.metrics import classification_report

import re

import string

#reading datasets

df_fake = pd.read_csv("Fake.csv")

df_true = pd.read_csv("True.csv")

df_fake.head()
```

	title	text	subject	date
0	Donald Trump Sends Out Embarrassing New Year'...	Donald Trump just couldn t wish all Americans ...	News	December 31, 2017
1	Drunk Bragging Trump Staffer Started Russian ...	House Intelligence Committee Chairman Devin Nu...	News	December 31, 2017
2	Sheriff David Clarke Becomes An Internet Joke...	On Friday, it was revealed that former Milwauk...	News	December 30, 2017
3	Trump Is So Obsessed He Even Has Obama's Name...	On Christmas day, Donald Trump announced that ...	News	December 29, 2017
4	Pope Francis Just Called Out Donald Trump Dur...	Pope Francis used his annual Christmas Day mes...	News	December 25, 2017

df_true.head()

	title	text	subject	date
0	As U.S. budget fight looms, Republicans flip t...	WASHINGTON (Reuters) - The head of a conservat...	politicsNews	December 31, 2017
1	U.S. military to accept transgender recruits o...	WASHINGTON (Reuters) - Transgender people will...	politicsNews	December 29, 2017
2	Senior U.S. Republican senator: 'Let Mr. Muell...	WASHINGTON (Reuters) - The special counsel inv...	politicsNews	December 31, 2017
3	FBI Russia probe helped by Australian diplomat...	WASHINGTON (Reuters) - Trump campaign adviser ...	politicsNews	December 30, 2017
4	Trump wants Postal Service to charge 'much mor...	SEATTLE/WASHINGTON (Reuters) - President Donal...	politicsNews	December 29, 2017

#inserting coloumn 'class' as a target feature

```
df_fake["class"] = 0

df_true["class"] = 1
```

Removing last 10 rows for manual testing

```
df_fake_manual_testing = df_fake.tail(10)

for i in range(23480,23470,-1):

    df_fake.drop([i], axis = 0, inplace = True)


df_true_manual_testing = df_true.tail(10)

for i in range(21416,21406,-1):

    df_true.drop([i], axis = 0, inplace = True)
```

df_fake_manual_testing.head(10)

	title	text	subject	date	class
23461	REPORT: 'Federal Government Escalated the Viol...	KILLED: Rancher and protest spokesman Robert ...	Middle-east	January 28, 2016	0
23462	BOILER ROOM – Oregon Standoff, Cuddle Parties,...	Tune in to the Alternate Current Radio Network...	Middle-east	January 28, 2016	0
23463	Eyewitness Says Feds Ambushed Bundys, 100 Shot...	Patrick Henningsen 21st Century Wire UPDATE: 1...	Middle-east	January 27, 2016	0
23464	Episode #119 – SUNDAY WIRE: 'You Know the Dril...	Episode #119 of SUNDAY WIRE SHOW finally resum...	Middle-east	January 24, 2016	0
23465	'There'll be boots on the ground': US making n...	21st Century Wire says Various parties in Wash...	Middle-east	January 23, 2016	0
23466	Boston Brakes? How to Hack a New Car With Your...	21st Century Wire says For those who still ref...	Middle-east	January 22, 2016	0
23467	Oregon Governor Says Feds 'Must Act' Against P...	21st Century Wire says So far, after nearly 20...	Middle-east	January 21, 2016	0
23468	Ron Paul on Burns Oregon Standoff and Jury Nul...	21st Century Wire says If you ve been followin...	Middle-east	January 21, 2016	0
23469	BOILER ROOM: As the Frogs Slowly Boil – EP #40	Tune in to the Alternate Current Radio Network...	Middle-east	January 20, 2016	0
23470	Arizona Rancher Protesting in Oregon is Target...	RTOne of the most visible members of the armed...	Middle-east	January 20, 2016	0

df_true_manual_testing.head(10)

	title	text	subject	date	class
21407	Mata Pires, owner of embattled Brazil builder ...	SAO PAULO (Reuters) - Cesar Mata Pires, the ow...	worldnews	August 22, 2017	1
21408	U.S., North Korea clash at U.N. forum over nuc...	GENEVA (Reuters) - North Korea and the United ...	worldnews	August 22, 2017	1
21409	U.S., North Korea clash at U.N. arms forum on ...	GENEVA (Reuters) - North Korea and the United ...	worldnews	August 22, 2017	1
21410	Headless torso could belong to submarine journ...	COPENHAGEN (Reuters) - Danish police said on T...	worldnews	August 22, 2017	1
21411	North Korea shipments to Syria chemical arms a...	UNITED NATIONS (Reuters) - Two North Korean sh...	worldnews	August 21, 2017	1
21412	'Fully committed' NATO backs new U.S. approach...	BRUSSELS (Reuters) - NATO allies on Tuesday we...	worldnews	August 22, 2017	1
21413	LexisNexis withdrew two products from Chinese ...	LONDON (Reuters) - LexisNexis, a provider of l...	worldnews	August 22, 2017	1
21414	Minsk cultural hub becomes haven from authorities	MINSK (Reuters) - In the shadow of disused Sov...	worldnews	August 22, 2017	1
21415	Vatican upbeat on possibility of Pope Francis ...	MOSCOW (Reuters) - Vatican Secretary of State ...	worldnews	August 22, 2017	1
21416	Indonesia to buy $1.14 billion worth of Russia...	JAKARTA (Reuters) - Indonesia will buy 11 Sukh...	worldnews	August 22, 2017	1

#merging both datasets

df_manual_testing = pd.concat([df_fake_manual_testing,df_true_manual_testing], axis = 0)

df_manual_testing.to_csv("manual_testing.csv")

df.head(10)

	title	text	subject	date	class
0	Donald Trump Sends Out Embarrassing New Year'...	Donald Trump just couldn t wish all Americans ...	News	December 31, 2017	0
1	Drunk Bragging Trump Staffer Started Russian ...	House Intelligence Committee Chairman Devin Nu...	News	December 31, 2017	0
2	Sheriff David Clarke Becomes An Internet Joke...	On Friday, it was revealed that former Milwauk...	News	December 30, 2017	0
3	Trump Is So Obsessed He Even Has Obama's Name...	On Christmas day, Donald Trump announced that ...	News	December 29, 2017	0
4	Pope Francis Just Called Out Donald Trump Dur...	Pope Francis used his annual Christmas Day mes...	News	December 25, 2017	0
5	Racist Alabama Cops Brutalize Black Boy While...	The number of cases of cops brutalizing and ki...	News	December 25, 2017	0
6	Fresh Off The Golf Course, Trump Lashes Out A...	Donald Trump spent a good portion of his day a...	News	December 23, 2017	0
7	Trump Said Some INSANELY Racist Stuff Inside ...	In the wake of yet another court decision that...	News	December 23, 2017	0
8	Former CIA Director Slams Trump Over UN Bully...	Many people have raised the alarm regarding th...	News	December 22, 2017	0
9	WATCH: Brand-New Pro-Trump Ad Features So Muc...	Just when you might have thought we d get a br...	News	December 21, 2017	0

#Keep only 2 columns(text,class)Remove others

```
df = df[["text", "class"]]
```

#**shuffling the data**

```
df = df.sample(frac=1).reset_index(drop=True)
```

```
df.head()
```

	text	class
10613	Well, that didn t take long Nancy Pelosi went ...	0
6606	(Reuters) - The following people are mentioned...	1
3002	(Reuters) - President Donald Trump's son-in-la...	1
18786	In a harsh reminder of just how differently re...	0
10808	RANCHO MIRAGE, Calif. (Reuters) - President Ba...	1

#text cleaning

```
def wordopt(text):
    text = text.lower()
    text = re.sub('\[.*?\]', '', text)
    text = re.sub("\\W"," ",text)
    text = re.sub('https?://\S+|www\.\S+', '', text)
```

```
    text = re.sub('<.*?>+', '', text)
    text = re.sub('[%s]' % re.escape(string.punctuation), '', text)
    text = re.sub('\n', '', text)
    text = re.sub('\w*\d\w*', '', text)
    return text
df["text"] = df["text"].apply(wordopt)
#Split data (x = input,y = output)
x = df["text"]
y = df["class"]
x_train, x_test, y_train, y_test = train_test_split(x, y, test_size=0.25)
vectorization = TfidfVectorizer()
xv_train = vectorization.fit_transform(x_train)
xv_test = vectorization.transform(x_test)
X_train, X_test, y_train, y_test = train_test_split(X, df["class"], test_size=0.25, random_state=42)
#Logistic regressor
model = LogisticRegression()
model.fit(xv_train,y_train)
y_pred=model.predict(xv_test)
# Accuracy
print("Accuracy:", round(accuracy_score(y_test, y_pred)*100,2), "%")
# Classification report
print(classification_report(y_test, y_pred))
```

OUTPUT

Accuracy: 98.49 %

	precision	recall	f1-score	support
0	0.99	0.98	0.99	5802
1	0.98	0.99	0.98	5423
accuracy			0.98	11225
macro avg	0.98	0.99	0.98	11225
weighted avg	0.98	0.98	0.98	11225

```
#model testing
def output_lable(n):
    if n == 0:
        return "Fake News"
    elif n == 1:
        return "Not A Fake News"

def manual_testing(news):
    testing_news = {"text":[news]}
    new_def_test = pd.DataFrame(testing_news)
    new_def_test["text"] = new_def_test["text"].apply(wordopt)
    new_x_test = new_def_test["text"]
    new_xv_test = vectorization.transform(new_x_test)
    y_pred = model.predict(new_xv_test)

    return print("\n\nLR Prediction: {}".format(output_lable(y_pred[0])))
```

news = str(input())

manual_testing(news)

OUTPUT

```
news = str(input())
manual_testing(news)
```

```
 In a harsh reminder of just how differently re..


LR Prediction: Fake News
```

```
news = str(input())
manual_testing(news)
```

```
 (Reuters) - The following people are mentioned...


LR Prediction: Not A Fake News
```

CONCLUSION

In this project, the Logistic Regression model was implemented to detect fake news based on textual features extracted from news articles. Logistic Regression proved to be highly effective for text classification because it performs well with high-dimensional data such as TF-IDF vectors. The model was able to learn patterns from the training dataset and accurately classify news articles as fake or real. It is computationally efficient, easy to implement, and provides stable results compared to many complex models. Logistic Regression also offers good generalization when applied to unseen data, making it suitable for real-world text classification tasks. In this project, the model achieved high accuracy, demonstrating its capability in identifying misleading information from news content. Therefore, Logistic Regression is a reliable and practical approach for fake news detection systems.